Gaussian Processes

- Gaussian Processes (`GPs`) allow us to make predictions about data by incorporating prior knowledge. For a given set of training points, there are potentially infinitely many functions that fit the data. `GPs` offer an elegant solution by assigning a probability to each of these functions. The mean of this probability distribution then represents the most probable characterization of the data. Furthermore, using a probabilistic approach allows us to incorporate the confidence of the prediction into the regression result.
- A random variable $\mathbf{x} \in \mathbb{R}^n$ follows a multivariate gaussian dist in n dimensions with mean $\mu \in \mathbb{R}^n$ and covariance $\Sigma \in \mathbb{R}^{n \times n}$ if:
$$p(\mathbf{x}; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\mu)^T \Sigma^{-1}(\mathbf{x}-\mu)\right)$$
- A `GP` is an extension of the multivariate gaussian to infinite dimensions. Given an input vector $x \in \mathbb{R}^n$, the `GP` returns a transformed vector $y \in \mathbb{R}^n$. Every component of y represents the *probability* of observing x_i according to some Gaussian living in dimension i.
- Since a GP acts as a probability distribution, we can also sample new y for new x apart from transforming an x into a y. Hence, if we sample the entire domain of x, we can obtain the function f such that $f(x) = y$.

Marginalization: Integrate along one of the dimensions of the Gaussian to get the probability distribution over the remaining dimensions:
$$P(X){=}\int_{-\infty}^{\infty} P(X,Y)dY \text{ or } P(X){=}\Sigma_Y P(X,Y)$$

Conditioning: Updates the distribution over function values at new points, given observations at known points. By conditioning on observed data, `GPs` allow us to effectively incorporate priors and make informed predictions in uncertain scenarios:
$$X|Y \sim \mathcal{N}(\mu_X + \Sigma_{XY}\Sigma_{YY}^{-1}(Y-\mu_Y), \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX})$$
$$Y|X \sim \mathcal{N}(\mu_Y + \Sigma_{YX}\Sigma_{XX}^{-1}(X-\mu_X), \Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY})$$
^^ Complex math, but note that the updated means only depend on the conditioned variable, while the covariance matrix is independent of the variable.

Functional view of `GPs`

- `GPs` (which are stochastic processes) are a set of random variables. Let each of these random variables have a corresponding index i, which represents the i^{th} dimension of the n-dimensional multivariate distribution.
- **Goal**: Learn the underlying distribution from *train data* (Y).
- `GPs` attempt to model the underlying distribution of the *test data* (X), by modeling the joint distribution $P_{X,Y}$ - the span of possible function values - as a multivariate normal distribution whose dimensionality is $|X|{+}|Y|$).
- Regressing on X requires modeling this as Bayesian inference, i.e., update the (current) hypothesis as information (Y) becomes available. This conditional probability ($P_{X|Y}$) is a multivariate normal distribution as well.

- `GP`s treat each test point as a random variable. Making predictions requires sampling from the distribution; the i^{th} component of the sampled vector is interpreted as the function value corresponding to the i^{th} test point.

Setting up the distribution requires defining μ and Σ. In `GP`s, centering the data, i.e., $\mu = 0$ is common practice. The covariance matrix (Σ) is determined by the `GP`'s covariance function k - the kernel of the `GP`.

Kernels

- The covariance matrix describes the shape of the distribution and determines the characteristics of the predicted function. Σ is determined by evaluating the kernel/ covariance function (k) pairwise on all the points and acts as a similarity measure:
$$k : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}; \Sigma = \mathrm{Cov}(X, X') = kt, t'$$
- Σ_{ij} describes how much influence the i^{th} and the j^{th} point have on each other. Since k describes the pairwise similarity between the function values, it controls the possible shape that a fitted function can adopt. This formulation allows the function to take on similarity measures beyond simple Euclidean distances.
- Stationary kernels such as the RBF kernel ($\sigma^2 \exp(-\frac{||t-t'||^2}{2l^2})$) or the periodic kernel ($\sigma^2 \exp(-\frac{2\sin^2(\pi|t-t'|/p)}{l^2})$) are invariant to translations and the covariance of two points is only dependent on their relative position. Non-stationary kernels such as the linear kernel ($\sigma_b^2 + \sigma^2(t-c)(t-c')$) do not have this constraint and depend on an absolute location.
- Different kernels can describe different classes of functions, which can be used to model the desired shape of the function.

Combining Kernels

- Kernels can be combined resulting in more specialized kernels, allowing domain experts to introduce priors to capture trends in the data.
- The covariance matrix of `GP`s has to be positive semi-definite. When choosing optimal kernel combinations, all methods that preserve this property are allowed (such as addition and multiplication).

Prior Distribution

- Training data is the *prior* distribution (P_Y). Centering the distribution (μ=0) is convention. We begin with μ=0 (dimensionality $|Y|$=N).
- The kernel function (dimensionality $N \times N$) determines which functions from the space of all possible functions are more probable.

- Adjusting the kernel function parameters controls the shape of the resulting functions and also varies the confidence of the predictions. For example, decreasing the variance (σ) results in sampled functions that are concentrated around the mean (μ).

Posterior Distribution

- Model the joint distribution ($P_{X,Y} \in \mathbb{R}^{|Y|+|X|} \times \mathbb{R}^{|Y|+|X|}$) and use train data ($Y$) to constrain the distribution and infer test data (X), i.e., the output is the conditional probability $P_{X|Y} \in \mathbb{R}^{|X|}$.
- Using conditioning, $P_{X|Y}$ can be derived from $P_{X,Y}$. Conditioning leads to derived versions of μ and Σ : $X|Y \sim \mathcal{N}(\mu', \Sigma')$ where $\mu' \neq 0$. This is because the training datapoints (Y) constrain the set of possible functions to those that pass through the training points.
- Further, precisely passing through Y may lead to overfitting/ over-complexity, especially with noisy real-world data. We model noise/ measurement errors with the error term $\epsilon \sim \mathcal{N}(0, \psi^2)$: $X = f(Y) + \epsilon$.
- The resulting joint probability distribution is:
$$P_{X,Y} = \begin{bmatrix} X \\ Y \end{bmatrix} \sim \mathcal{N}(0, \Sigma) = \mathcal{N}(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} + \psi^2 I \end{bmatrix})$$
- We obtain obtain a prediction for our function values by sampling from the conditional distribution which is derived from conditioning on the joint distribution described above.
- Since sampling involves randomness, the prediction may not be deterministic. Marginalizing (the noise) associated with each random variable (i) allows us to extract the function value (μ'_i) and uncertainty ($\sigma'_i = \Sigma'_{ii}$) for the i^{th} test point.

References

1. Gaussian Processes - Reid Pryzant.
2. A Visual Exploration of Gaussian Processes - Jochen Görtler, Rebecca Kehlbeck, and Oliver Deussen.
3. Gaussian Processes, not for dummies - Yuge Shi.